

EXPERIMENT 20. Information Retrieval using TF-IDF

Aim: To implement a basic information retrieval system using TF-IDF for document ranking.

Program:

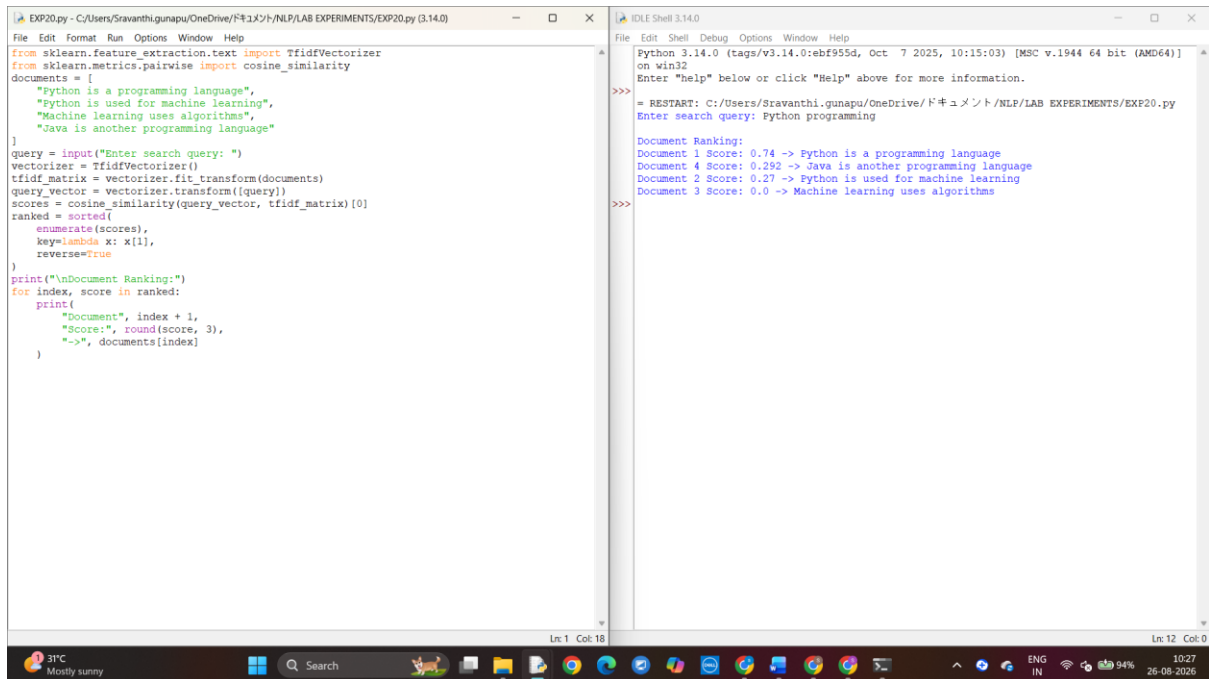
```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

documents = [
    "Python is a programming language",
    "Python is used for machine learning",
    "Machine learning uses algorithms",
    "Java is another programming language"
]

query = input("Enter search query: ")
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(documents)
query_vector = vectorizer.transform([query])
scores = cosine_similarity(query_vector, tfidf_matrix)[0]
ranked = sorted(
    enumerate(scores),
    key=lambda x: x[1],
    reverse=True
)

print("\nDocument Ranking:")
for index, score in ranked:
    print(
        "Document", index + 1,
        "Score:", round(score, 3),
        "->", documents[index]
    )
```

Output:



The screenshot shows a Python script in a file editor and its execution output in the IDLE Shell. The script uses TF-IDF vectorization and cosine similarity to rank documents based on a search query. The output shows the document ranking and scores.

```
EXP20.py - C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP20.py (3.14.0)
File Edit Format Run Options Window Help
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
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IDLE Shell 3.14.0
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP20.py
Enter search query: Python programming
Document Ranking:
Document 1 Score: 0.74 -> Python is a programming language
Document 4 Score: 0.292 -> Java is another programming language
Document 2 Score: 0.27 -> Python is used for machine learning
Document 3 Score: 0.0 -> Machine learning uses algorithms
>>>
```

31°C Mostly sunny 10:27 26-08-2026